

Advanced Praktika SoSe 26 Notes

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Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.

Part I

Notes

1 Project Setup

1.1 General repository structure

The whole project is managed inside a Quarto book repository. The Quarto project contains notes, theory summaries, the final report, and the implementation code.

The planned directory structure is:

```
project-root/  
  index.qmd  
  report/  
  notes/  
  code/  
    CMakeLists.txt  
    external/  
      hdnum/          # git submodule  
        cpfloat/      # cloned inside hdnum  
    include/  
      mixed_ir.hpp     # algorithm  
      hdnum_conversions.hpp # convert()  
      error_metrics.hpp # forward/backward error helpers, later  
    experiments/  
      exp_convergence.cc  
      exp_condition_sweep.cc  
      exp_residual_precision.cc  
    examples/  
      hdnum_sandbox.cc  
    tests/  
      test_conversion.cc  
      test_mixed_ir_sanity.cc  
    scripts/  
      run_all_experiments.py  
      plot_results.py  
    results/  
      raw/  
      plots/
```

Purpose of the main sub-directories:

external/	external dependencies, especially HDNUM
include/	own header-only algorithm and helper code
src/	optional implementation files, if needed later
examples/	small sandbox/demo programs for learning and testing
experiments/	reproducible experimental drivers for report data
tests/	correctness and sanity checks
scripts/	automation and plotting scripts
results/raw/	generated CSV or data files
results/plots/	generated figures for the report

1.1.1 Experimental Drivers

They will go in:

```
code/  
  experiments/  
    exp_convergence.cc  
    exp_condition_sweep.cc  
    exp_residual_precision.cc
```

They will be written in c++ because they need to instantiate HDNUM types like FP16, bfloat16, FP64, FP128, FP256, run LU, perform refinement, and compute errors.

But the overall experiment workflow will be split:

```
C++ drivers:  
  run numerical experiments  
  write CSV files  
  
Python / shell scripts:  
  run many driver configurations  
  collect results  
  make plots
```

like:

```
code/
  experiments/
    exp_convergence.cc
    exp_condition_sweep.cc
    exp_residual_precision.cc
  scripts/
    run_all_experiments.py
    plot_results.py
  results/
    raw/
    plots/
```

The cpp experiments can be written so that they can be run like:

```
./build/exp_condition_sweep --uf FP16 --u FP64 --ur FP128 --n 100 --matrix random_orthogonal
./build/exp_condition_sweep --uf bfloat16 --u FP64 --ur FP128 --n 100 --matrix random_orthogonal
./build/exp_condition_sweep --uf FP32 --u FP64 --ur FP128 --n 100 --matrix random_orthogonal
```

They a python or shell script can run all of those automatically.

Precision sweeps should vary precision, condition number, and matrix type, and experiments should be scripted so rerunning them is convenient. In practice, that means your experiment system should let you rerun a whole table of experiments with one command, rather than manually recompiling/editing files.

1.2 HDNUM, CPFloat and GMP

HDNUM is included as a git submodule:

```
git submodule add https://parcomp-git.iwr.uni-heidelberg.de/Teaching/hdnum.git code/external/hdnum
git submodule update --init --recursive
```

CPFloat is cloned and built inside the HDNUM top-level directory, matching HDNUM's expected layout:

```
cd code/external/hdnum
git clone https://github.com/north-numerical-computing/cpfloat.git
cd cpfloat
make lib
```


GMP is installed system-wide:

```
sudo apt install libgmp-dev
```

Because CPFloat sits inside the HDNUM submodule but is not part of HDNUM itself, untracked files are hidden locally only for that submodule:

```
cd code/external/hdnum
git config status.showUntrackedFiles no
```

and the parent repository ignores untracked content specifically inside the HDNUM submodule:

```
git config submodule.code/external/hdnum.ignore untracked
```

1.3 Changing Top-level CMakeLists.txt

Currently the implementations are header-only and are contained in the folder include. Later if source files that not only header are added e.g. to a src/ folder it has to be accounted for:

Right now, this part:

```
add_library(mixed_precision_core INTERFACE)
```

is only an INTERFACE target. That means it does not compile any .cc files. It only carries usage requirements:

```
include_directories
compile_definitions
linked_libraries
compile_features
```

So it works well for a header-only project, but it does not yet build files from src/.

But if later you add:

```
src/csv_writer.cc
src/matrix_generators.cc
src/experiment_io.cc
```

then those files will not automatically be compiled unless we update CMake.

1.3.1 How to integrate src/ later

The usual clean solution is to change from a purely INTERFACE core target to a real library target.

For example:

```
add_library(mixed_precision_core
    src/csv_writer.cc
    src/matrix_generators.cc
    src/experiment_io.cc
)
```

Then keep the same include directories and dependencies:

```
target_include_directories(mixed_precision_core
    PUBLIC
        "${CMAKE_CURRENT_SOURCE_DIR}/include"
        "${HDNUM_DIR}"
        "${CPFLOAT_INCLUDE_DIR}"
)

target_compile_definitions(mixed_precision_core
    PUBLIC
        HDNUM_HAS_GMP=1
        HDNUM_HAS_CPFLOAT=1
)

target_link_libraries(mixed_precision_core
    PUBLIC
        "${CPFLOAT_LIBRARY}"
        gmpxx
        gmp
)

target_compile_features(mixed_precision_core
    PUBLIC
        cxx_std_17
)
```

Then all your examples, tests, and experiments can continue doing:

```
target_link_libraries(my_target
    PRIVATE
        mixed_precision_core
        mixed_precision_warnings
)
```

and they will automatically get both:

```
the headers from include/
the compiled implementation files from src/
```

1.3.2 A flexible version

If you do not want to manually list every source file, you can use globbing:

```
file(GLOB project_sources CONFIGURE_DEPENDS
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cc"
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cpp"
)

add_library(mixed_precision_core
    ${project_sources}
)
```

But there is one small issue: `add_library()` with an empty source list can be awkward in some CMake versions/workflows.

So for now, while `src/` is empty, I would keep the current `INTERFACE` version.

Later, when you add the first real `.cc` file, change:

```
add_library(mixed_precision_core INTERFACE)
```

to:

```
file(GLOB project_sources CONFIGURE_DEPENDS
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cc"
    "${CMAKE_CURRENT_SOURCE_DIR}/src/*.cpp"
)

add_library(mixed_precision_core
```

```
    ${project_sources}  
)
```

and replace INTERFACE with PUBLIC in the relevant `target_*` commands.

1.3.3 Rule of thumb

Use this now:

```
add_library(mixed_precision_core INTERFACE)
```

while the project is header-only.

Use this later:

```
add_library(mixed_precision_core ${project_sources})
```

once `src/` contains real compiled support code.

2 Project Summary

2.1 Abstract

This *Fortgeschrittenenpraktikum* project investigates mixed precision iterative refinement for solving linear systems

$$Ax = b.$$

The goal is to implement a three-precision LU-based method: the LU factorization and correction solves use low precision u_f , the solution updates use working precision u , and the residuals are computed in higher precision u_r . Starting from a low-precision LU solve, the algorithm repeatedly computes $r_i = b - Ax_i$, solves for a correction d_i , and updates x_i until $|d_i|/|x_i| < u$ or a maximum iteration count is reached.

The focus is empirical accuracy, not runtime, since low precisions such as fp8, fp16, and bfloat16 are simulated via CPFloat. The experiments will study convergence, forward/backward error, condition-number dependence, precision triples (u_f, u, u_r) residual precision, and comparison with a direct working-precision solve.

Summary:

- implement three-precision variant of **iterative refinement for linear systems**
- study its accuracy

2.1.1 Problem

- A linear system $Ax = b$. **iterative refinement** improves an approximate solution by repeatedly computing the residual and solving a correction equation.
 - LU factorization in low precision: u_f
 - **working arithmetic** in ‘middle’ precision: u
 - residual computation in high precision: u_r

i Note

Review the definition of **accuracy** of an algorithm

- precision triples (u_r, u, u_f) with $u_f \succ u \succ u_r$:
 - i.e. u_f low precisions like: **fp8**, **fp16**, **bfloat16**, come from CPFloat:
 - * FP8: CPFloat<4, 4>
 - * bfloat16: CPFloat<8, 8>
 - * FP16: CPFloat<11, 5>
 - u is probably the native FP64 i.e. double precision

i Note

Study the FP8, FP16 and bfloat16 formats and how they relate to template parameters in CPfloat<>

2.1.2 Algorithm

Algorithm: Three-precision iterative refinement

1. Compute the LU factorization $PA = LU$ in precision u_f .
2. Solve $LU x_0 = P b$ in precision u_f ; cast x_0 to precision u .
3. While not converged:
 4. Compute $r_i = b - A x_i$ in precision u_r .
 5. Cast r_i to precision u .
 6. Solve $LU d_i = P r_i$ in precision u_f ; cast d_i to precision u .
 7. Update $x_{i+1} = x_i + d_i$ in precision u .

- convergence condition: $\frac{\|d_i\|}{\|x_i\|} < u$ where u is the unit roundoff of the working precision,
- or stop after a max number of iterations.

i Note

why is $\frac{\|d_i\|}{\|x_i\|} < u$ the convergence condition?

2.1.3 HDNUM & Own Implementation

relevant parts:

matrix and vector classes:

- `src/densematrix.hh`: `DenseMatrix<T>` with:
 - `mv`: matrix vector product
 - `mm`: matrix matrix product
 - `transpose()` etc
- `src/vector.hh`: `Vector<T>` with vector arithmetic operations (multiplication with scalar, addition) and norms
- LU decomposition in `src/lr.hh`:
 - `lr_fullpivot(A, p, q)`
 - `lr_partialpivot(A, p)`
 - triangular solves: `solveL(A, x, b)`, `solveR(A, x, b)`
 - `permute_forward` and `permute_backward`

2.1.4 Mixing Types

In a mixed precision algorithm different parts of the algorithm are instantiated / computed with different precisions, i.e. different types:

- LU factorization is done in lower precision u_f : E.g. `DenseMatrix<FP16>`
- Residual computation in higher precision u_r : E.g. `DenseMatrix<FP128>`

Therefore we must be able to convert from one type (class) like `DenseMatrix<FP16>` to another type (class) like `DenseMatrix<FP128>`. This conversion requires explicit construction or element-wise copying.

Managing type conversions between precisions is they key challenge. It is suggested to do this with a explicit template functions for matrices and vectors:

```
template<class T_out, class T_in>
void convert(Vector<T_out>& out, const Vector<T_in>& in)

template<class T_out, class T_in>
void convert(DenseMatrix<T_out>& out, const Vector<T_in>& in)
```

2.1.5 Methodology

- **reference solutions**: run a standard solver for $A \cdot x = b$ in FP256 and treat the result x_{ex} as exact.
- **error metrics**:
 - report errors using norms appropriate to your algorithm

- compute **all error measures** in a precision higher than the working precision of your algorithm, i.e. u
- otherwise you are measuring the precision of your measurement.

i Note

- What does **error** mean in this context? $\|x^* - x_{ex}\|_2$?
- What is a norm that is **appropriate** to LU, simply $\|\bullet\|_2$?
- Does Compute error measures in a precision higher than u simply mean that we compute $\|\cdot\|$ in something like FP128
 - what does “otherwise you are measuring the precision of your measurement” mean?

- experiments should be scripted so that re-running them with various precision, condition number and matrix type is convenient.

2.1.6 What Should be Studied

- **Convergence Histories:** for each precision combination (u_f, u, u_r) and several test matrices A_i with different condition numbers $k(A_i)$:
 - plot **forward error** as a function of i
 - plot **backward error** as a function of i

i Note

review the definitions of **forward error** and **backward error**

- **Summary Across Condition Numbers:** for a wide range of $k(A)$ for each precision combinations
 - record:
 - * the final achieved **forward error**
 - * number of iterations
 - plot these as a function of $k(A)$: theory predicts convergence requires roughly $k(A) \cdot u_f < 1$.
- **The Role u_r :** Fix (u_f, u) and vary u_r . When does $u_r > u$ make a measurable difference

i Note

- Normally $u_r < u$, i.e. the residual computation is done in a precision higher than the working precision.
- What happens when we lower the precision, when does it make a difference?

2.1.7 Plan

1. Implement `hdnum_conversions.hpp`.
2. Add conversion tests.
3. Implement a minimal `mixed_ir.hpp`.
4. First test with all precisions set to FP64.
5. Then test real mixed precision triples.
6. Add systematic experiments and plotting scripts.

3 Conversions

3.1 Floating Point Types

In this project there are three groups of number formats:

1. simulated low precision via CPFloat
2. native C++ hardware types `float` and `double`
3. high / arbitrary precision via GMP

The low precision `fp8`, `fp16`, `bfloat16` are simulated via CPFloat.

An overview of types defined in HDNUM:

HDNUM / project type	Provider	Parameters	Corresponds to in Higham–Mary	Unit roundoff, roughly
FP8	CPFloat	CPFloat<4,4>	custom 8-bit / “quarter precision”-like format	$2^{-4} \approx 6.25 \cdot 10^{-2}$
bfloat16	CPFloat	CPFloat<8,8>	Higham–Mary <code>bfloat16</code>	$2^{-8} \approx 3.91 \cdot 10^{-3}$
FP16	CPFloat	CPFloat<11,5>	IEEE half precision, Higham–Mary <code>fp16</code>	$2^{-11} \approx 4.88 \cdot 10^{-4}$
FP32	native C++ <code>float</code>	effectively (24,8)	IEEE single precision, Higham–Mary <code>fp32</code>	$2^{-24} \approx 5.96 \cdot 10^{-8}$
FP64	native C++ <code>double</code>	effectively (53,11)	IEEE double precision, Higham–Mary <code>fp64</code>	$2^{-53} \approx 1.11 \cdot 10^{-16}$
FP128	GMP	approx. 128-bit, FP<64>	software high precision / “quadruple-like” role, not necessarily IEEE <code>fp128</code>	depends on HDNUM wrapper
FP256	GMP	approx. 256-bit, FP<192>	multiprecision reference precision	depends on chosen GMP precision

HDNUM / project type	Provider	Parameters	Corresponds to in Higham–Mary	Unit roundoff, roughly
FP512, FP1024	GMP	approx. 512 / 1024-bit	multiprecision, beyond Higham–Mary’s standard table	depends on chosen GMP precision

As a reminder the three-precision LU-IR algorithm is:

1. Compute the LU factorization $PA = LU$ in precision u_f .
2. Solve $LU x_0 = P b$ in precision u_f ; cast x_0 to precision u .
3. While not converged:
 4. Compute $r_i = b - A x_i$ in precision u_r .
 5. Cast r_i to precision u .
 6. Solve $LU d_i = P r_i$ in precision u_f ; cast d_i to precision u .
 7. Update $x_{i+1} = x_i + d_i$ in precision u .

i.e.:

- 1) Compute LU in u_f
- 2) Solve the initial system in u_f
- 3) Compute residuals in u_r
- 4) Cast residuals to u
- 5) Solve correction equations using the low-precision LU factors in u_f
- 6) update in u

For the algorithm the intended mapping is:

Algorithm role	Meaning	Typical project choices
u_f	factorization precision; low precision used for LU	FP8, bfloat16, FP16, sometimes FP32
u	working precision; stores A, b, x_i and updates solution	usually FP32 or FP64
u_r	residual precision; computes $r_i = b - Ax_i$ accurately	FP64, FP128, maybe FP256 for experiments

A reasonable set of experiments could be:

Experiment	u_f	u	u_r
baseline sanity check	FP64	FP64	FP64
classical mixed precision	FP32	FP64	FP64
half/double refinement	FP16	FP64	FP64
half/double with higher residual	FP16	FP64	FP128
bfloat comparison	bfloat16	FP64	FP64
very low precision stress test	FP8	FP64	FP128

3.2 Type Conversions

Again, the precisions for the algorithm are:

- 1) u_f : factorization / solve precision (low)
- 2) u : working precision (medium)
- 3) u_r : residual precision (high)

LU is computed in u_f , the initial solution is cast to u , the residual is computed in u_r and cast back to u , the correction solve uses the low-precision LU factors, and the correction is cast to u before updating.

So the practically necessary conversions are:

Conversion		
$u \rightarrow u_f$		create low-precision copies of A , b , or residuals for LU / triangular solves.
$u_f \rightarrow u$		bring x_0 and correction d_i back to working precision.
$u \rightarrow u_r$		compute $r_i = b - Ax_i$ in high precision.
$u_r \rightarrow u$		The residual is cast back before solving the correction equation.
$u_f \leftrightarrow u_r$	not strictly needed	You normally convert through the working representation, or independently convert from the original A, b, x .

3.2.1 Implementing Conversions - Naive Approach

HDNum implements wrappers `FP` and `CPFloat` for the high-precision GMP and the low precision `CPFloat` types in the source files `highprec_gmp.hh` and `lowprec_cpfloat.hh`.

We are asked not to modify HDNum source files, instead implement explicit conversion functions:

```

template<class T_out, class T_in>
void convert(Vector<T_out>& out, const Vector<T_in>& in);

template<class T_out, class T_in>
void convert(DenseMatrix<T_out>& out, const DenseMatrix<T_in>& in);

```

The approach is to assign to `out` element-wise in loops, where the values constructed from the corresponding elements of `in`. A naive approach is to do:

```

#pragma once

#include <cstddef>
#include "hdnum.hh"

namespace mpir {

// Vector conversion: resizes output if needed.
template<class T_out, class T_in>
void convert(hdnum::Vector<T_out>& out,
             const hdnum::Vector<T_in>& in)
{
    if (out.size() != in.size()) {
        out.resize(in.size());
    }

    for (std::size_t i = 0; i < in.size(); ++i) {
        out[i] = T_out(in[i]);
    }
}

// Convenience wrapper: allocates a new vector.
template<class T_out, class T_in>
hdnum::Vector<T_out> convert_vector(const hdnum::Vector<T_in>& in)
{
    hdnum::Vector<T_out> out(in.size());
    convert(out, in);
    return out;
}

// Matrix conversion: output matrix must already have correct shape.
template<class T_out, class T_in>

```

```

void convert(hdnum::DenseMatrix<T_out>& out,
             const hdnum::DenseMatrix<T_in>& in)
{
    if (out.rowsize() != in.rowsize() || out.colsize() != in.colsize()) {
        throw std::invalid_argument(
            "mpir::convert(DenseMatrix): output matrix has wrong size"
        );
    }

    for (std::size_t i = 0; i < in.rowsize(); ++i) {
        for (std::size_t j = 0; j < in.colsize(); ++j) {
            out[i][j] = T_out(in[i][j]);
        }
    }
}

// Convenience wrapper: allocates a new matrix.
template<class T_out, class T_in>
hdnum::DenseMatrix<T_out> convert_matrix(const hdnum::DenseMatrix<T_in>& in)
{
    hdnum::DenseMatrix<T_out> out(in.rowsize(), in.colsize());
    convert(out, in);
    return out;
}

```

Where the type conversion is done via the assignment:

```
out[i] = T_out(in[i])
```

for each element.

But this doesn't work for all pairs of `T_in`, `T_out` because not all the necessary constructors or conversion operators are implemented in the classes `CPFloat` and `FP`

Which Conversions Work which Don't ?

FP provides the constructors:

- `FP()`
- `FP(const double &)`
- `FP(const float &)`
- `FP(const int &)`

- `FP(const mpf_class &)`
- `FP<mm>(const FP<mm>&)`

So the conversion:

- `double ==> GMP`

works

`CPFloat` provides the constructors:

- `CPFloat()`
- `CPFloat(const double &)`
- `CPFloat<mm, ee>(const double &)`

So:

- `double ==> CPFloat`

works

`CPFloat` also provides the conversion operator `operator double() const`. Therefore

- `CPFloat ==> double`

works as well.

Because `CPFloat` can be converted to `double` and `GMP` can be constructed from a `double`:

- `CPFloat ==> GMP`

works too, by the implicit conversion of `CPFloat` to a `double`

But since the `FP` class doesn't provide a `double()` operator the conversion

- `GMP ==> double`

doesn't work.

Also `CPFloat` doesn't implement a constructor that accepts a `GMP` type, (or also because `FP` doesn't provide a `double()` operator) the conversion:

- `GMP ==> CPFloat`

doesn't work.

We summarize this with the following diagram and table:

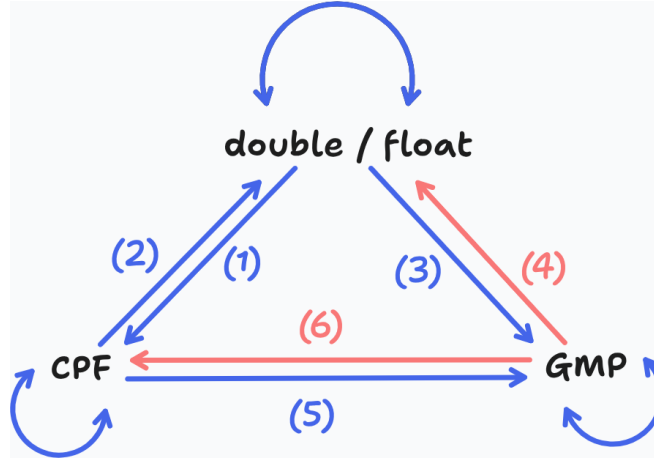


Figure 3.1: conversion triangle

Table 3.5: conversion table

Nr	Conversion	Works	
		?	Why ?
(1)	<code>double</code> $\Rightarrow$ <code>CPFloat</code>	Yes	<code>CPFloat(const double &)</code>
(2)	<code>CPFloat</code> $\Rightarrow$ <code>double</code>	Yes	<code>CPFloat::double()</code>
(3)	<code>double</code> $\Rightarrow$ <code>GMP</code>	Yes	<code>FP(const double &)</code>
(4)	<code>GMP</code> $\Rightarrow$ <code>double</code>	No	No <code>FP::double()</code>
(5)	<code>CPFloat</code> $\Rightarrow$ <code>GMP</code>	Yes	<code>CPFloat</code> is converted to <code>double</code> implicitly
(6)	<code>GMP</code> $\Rightarrow$ <code>CPFloat</code>	No	No <code>FP(const CPFloat &)</code> or no <code>FP::double()</code>

So if the conversion operator `FP::double()` would be implemented both (4) and (6) would work. (6) would work too because `GMP` type would be implicitly converted to a `double` and a `CPFloat` would be constructed. But more explicitly a constructor `CPFloat(const FP &)` can be implemented too.

But since we are initially not allowed to modify `HDNUM` source code we take a different approach.

3.2.2 Structure of FP and CPFloat Classes

FP	CPFloat
-mpf_class number	-double number
+FP() +FP(const double &) +FP(const mpf_class &) +FP(...) +getNumber() : mpf_class	+CPFloat() +CPFloat(const double &) +CPFloat(...) +operator double() +getNumber() : double

- FP:
 - FP has a single private member variable `number` that is an GMP type, i.e. `mpf_class`.
 - it can be accessed with the public getter `const getNumber() : mpf_class`
- CPFloat:
 - CPFloat has a single private member variable `number` of type `double`.
 - it can be accessed with the public getter `const getNumber() : double`

Note

There's a mistake in HDNUM `highprec_gmp.hh` source file. The member variable `number` is documented as:

Listing 3.1 `highprec_gmp.hh`

```
private:  
    // stores a double to represent the low precision number  
    mpf_class number;
```

The comment `// stores a double to represent the low precision number` is wrong, and is probably copied over by mistake from `lowprec_cpfloat.hh`:

Listing 3.2 lowprec_cpfloat.hh

```
private:
    // stores a double to represent the low precision number
    double number;
```

Instead it should be something like:

```
// stores an mpf_class to represent the high precision GMP number
```

With the above interface, given a GMP number, the double representation of the number can be obtained with:

Listing 3.3 get_d() method

```
hdnum::FP128 fp128_number(1.2345);
auto d = fp128_number.getNumber().get_d(); //d is double
```

Where we use the `.get_d()` method of `mpf_class` to get the `double` representation of the GMP number.

3.2.3 Implementing Conversions - Correct Approach

The main issue is that

```
out[i] = T_out(in[i]);
```

does not work uniformly for all type pairs. We replace this direct construction step with:

```
out[i] = scalar_convert<T_out>(in[i]);
```

This selects the appropriate conversion path at compile time, depending on the input and output types using standard library type traits and `if constexpr`

As summarized in Figure 3.1 and Table 3.5, the only conversions that do not work by direct construction are:

1. FP $\Rightarrow$ double,
2. FP $\Rightarrow$ CPFloat

All other relevant cases can be handled by direct construction of the form:

```
T_out(x) // where x has type T_in
```

In C++, we can check at compile time whether an object of type `T_out` can be constructed from an argument type `const T_in&` using the standard library type trait:

```
std::is_constructible_v<T_out, const T_in&>
```

This expression evaluates to `true` if such a construction is valid, and to `false` otherwise.

It can be verified that this indeed evaluates to `false` only for the conversion pairs:

- `T_out == double, T_in == FP`
- `T_out == CPFloat, T_in == FP`

Then, the initial part of `scalar_cast` can be written as:

```
template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x); // we use the direct construction
    }
    else { //remaining cases: FP => double || FP => CPFloat
        ...
    }
}
```

Logically, we know that the only possible cases in the `else` branch are:

- `T_out == double, T_in == FP`
- `T_out == CPFloat, T_in == FP`

In both cases `T_in` is an FP type. as we've seen in Listing 3.3 the `double` representation of an FP number `x` can be extracted with:

```
x.getNumber().get_d()
```

Since both `CPFloat` and `double` can be constructed from a `double`, we could simply write:

```

template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x); // we use the direct construction
    }
    else { //remaining cases: FP => double || FP => CPFloat
        return T_out(x.getNumber().get_d());
    }
}

```

However, this version is not robust enough because the `else` branch relies on our prior reasoning that only the two exceptional cases can reach it. The code itself does not express this assumption. If `scalar_cast` is later instantiated with another unsupported conversion pair, the compiler will still try to compile

```
x.getNumber().get_d()
```

even if `T_in` is not an FP type. This should lead to implementation-dependent and potentially confusing error message.

Therefore, it is better to make the remaining cases explicit using another `if constexpr` condition:

```

template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x); // direct construction works
    }
    else if constexpr (
        is_hdnum_fp_v<T_in> &&
        std::is_constructible_v<T_out, double>
    ) {
        return T_out(x.getNumber().get_d());
    }
    else {
        static_assert(always_false<T_out, T_in>::value,
            "unsupported scalar conversion");
    }
}

```

Here, the helper trait `is_hdnum_fp_v<T>` detects whether a type `T` is one of HDNUM's GMP-based high precision number types `hdnum::FP<m>`. This is a compile-time Boolean value, and analogous to the standard library traits such as `std::is_constructible_v`.

The implementation, given below, follows a common template metaprogramming partial specialization pattern:

```
// Detect HDNUM's GMP wrapper type hdnum::FP<m>
template<class T>
struct is_hdnum_fp_impl : std::false_type {};

template<int m>
struct is_hdnum_fp_impl<hdnum::FP<m>> : std::true_type {};

template<class T>
inline constexpr bool is_hdnum_fp_v =
    is_hdnum_fp_impl<
        std::remove_cv_t<std::remove_reference_t<T>>
    >::value;
```

1) The first part

```
template<class T>
struct is_hdnum_fp_impl : std::false_type {};
```

is the default case and says:

for an arbitrary type `T`, it is false

So:

```
is_hdnum_fp_impl<double>::value      // false
is_hdnum_fp_impl<float>::value        // false
is_hdnum_fp_impl<hdnum::FP16>::value // false, because CPFloat, not GMP FP
```

2) The second part

```
template<int m>
struct is_hdnum_fp_impl<hdnum::FP<m>> : std::true_type {};
```

is the specialization case, and says:

if the type has the form `hdnum::FP<m>`, it is true.

So:

```
is_hdnum_fp_impl<hdnum::FP128>::value // true
is_hdnum_fp_impl<hdnum::FP256>::value // true
```

3) The last part

```
template<class T>
inline constexpr bool is_hdnum_fp_v =
    is_hdnum_fp_impl<
        std::remove_cv_t<std::remove_reference_t<T>>
    >::value;
```

defines a compile-time boolean variable template. For example:

```
is_hdnum_fp_v<double> // false
is_hdnum_fp_v<hdnum::FP128> // true
```

It is analogous to standard library helpers such as:

```
std::is_arithmetic_v<T>
std::is_constructible_v<T, Args...>
std::is_convertible_v<From, To>
```

and allows to write

```
is_hdnum_fp_v<T>
```

instead of

```
is_hdnum_fp_impl<T>::value
```

The keyword `inline` is needed for variable that are defined in header files.

The use of:

```
std::remove_reference_t<T>
std::remove_cv_t<T>
```

makes the trait insensitive to references and `const` / `volatile` qualifiers. For example this way all of the following are treated as the same:

```
hdnum::FP<64>
const hdnum::FP<64>
hdnum::FP<64>&
const hdnum::FP<64>&
```

Without removing references and cv-qualifiers, the specialization would not match types such as `hdnum::FP<m>&`

The final branch of `scalar_cast`

```
else{
    static_assert(always_false<T_out, T_in>::value,
        "unsupported scalar conversion");
}
```

This branch is reached only if neither of the previous compile-time conditions applies:

```
std::is_constructible_v<T_out, const T_in&>
```

is `false`, so direct construction does not work, and

```
is_hdnum_fp_v<T_in> && std::is_constructible_v<T_out, double>
```

is also `false`, so the special `FP => double / FP => CPFloat` conversion path doesn't apply either.

Therefore, if the compiler reaches this final 'else' branch, the requested scalar conversion is unsupported. In that case, we want compilation to fail with a clear error message.

The helper trait is defined as:

```
template<class...>
struct always_false : std::false_type {};
```

This defines a type trait that is always `false`, regardless of the template arguments. For example:

```
always_false<int>::value           // false
always_false<double, int>::value   // false
always_false<hdnum::FP<64>, double>::value // false
```

The `class...` syntax denotes a template parameter pack and means that `always_false` can accept any number of template type arguments. In our case we use two arguments:

```
always_false<T_out, T_in>::value
```

It is important that `always_false<T_out, T_in>::value` depends on template parameters `T_out` and `T_in`, as it lets us create a compile-time `false` value that is evaluated when the templates are instantiated and not when the function is parsed. Omitting and writing simply `static_assert(false, ...)` would cause the assertion to be checked when the function template is parsed, and not only when the template is instantiated and the final branch is actually selected for the concrete instantiation.

scalar_cast Final Version

```
#pragma once

#include <cstdint>
#include <stdexcept>
#include <type_traits>

#include "hnum.hh"

namespace mpir {

// Helper for dependent static_assert(false)
template<class...>
struct always_false : std::false_type {};

// Detect HDNUM's GMP wrapper type hnum::FP<m>
template<class T>
struct is_hnum_fp_impl : std::false_type {};

template<int m>
struct is_hnum_fp_impl<hnum::FP<m>> : std::true_type {};

template<class T>
inline constexpr bool is_hnum_fp_v =
    is_hnum_fp_impl<
        std::remove_cv_t<std::remove_reference_t<T>>
    >::value;

// Scalar conversion
template<class T_out, class T_in>
T_out scalar_cast(const T_in& x)
{
    if constexpr (std::is_constructible_v<T_out, const T_in&>) {
        return T_out(x);
    }
    else if constexpr (
        is_hnum_fp_v<T_in> &&
        std::is_constructible_v<T_out, double>
    ) {
```



```

        return T_out(x.getNumber().get_d());
    }
    else {
        static_assert(always_false<T_out, T_in>::value,
                      "mpir::scalar_cast: unsupported scalar conversion");
    }
}

// Vector conversion
template<class T_out, class T_in>
void convert(hdnum::Vector<T_out>& out,
             const hdnum::Vector<T_in>& in)
{
    if (out.size() != in.size()) {
        throw std::invalid_argument(
            "mpir::convert(Vector): output vector has wrong size"
        );
    }

    for (std::size_t i = 0; i < in.size(); ++i) {
        out[i] = scalar_cast<T_out>(in[i]);
    }
}

// DenseMatrix conversion
template<class T_out, class T_in>
void convert(hdnum::DenseMatrix<T_out>& out,
             const hdnum::DenseMatrix<T_in>& in)
{
    if (out.rowsize() != in.rowsize() || out.colsize() != in.colsize()) {
        throw std::invalid_argument(
            "mpir::convert(DenseMatrix): output matrix has wrong size"
        );
    }

    for (std::size_t i = 0; i < in.rowsize(); ++i) {
        for (std::size_t j = 0; j < in.colsize(); ++j) {
            out[i][j] = scalar_cast<T_out>(in[i][j]);
        }
    }
}

```

```

    }
}

// convenience function
template<class T_out, class T_in>
hdnum::DenseMatrix<T_out> convert_matrix(const hdnum::DenseMatrix<T_in>& in)
{
    hdnum::DenseMatrix<T_out> out(in.rowsize(), in.colsize());
    convert(out, in);
    return out;
}

// convenience function
template<class T_out, class T_in>
hdnum::Vector<T_out> convert_vector(const hdnum::Vector<T_in>& in)
{
    hdnum::Vector<T_out> out(in.size());
    convert(out, in);
    return out;
}

} // namespace mpir

```

3.3 Testing `hdnum_conversions.hpp`

The file `test_conversion.cc` tests the conversion infrastructure implemented in `hdnum_conversions.hpp`. Its goals are to test that:

1. `scalar_cast<T_out>(x)` compiles for representative type.
2. `scalar_cast<T_out>(x)` returns the requested output type.
3. lossy low-precision conversions visibly lose information.
4. `convert(Vector)` applies `scalar_cast` element-wise correctly.
5. `convert(DenseMatrix)` applies `scalar_cast` elementwise correctly.
6. Vector and Matrix sizes are handled correctly.
7. Wrong-sized output containers throw exceptions
8. The convenience functions `convert_vector` and `convert_matrix` allocate correctly.

3.3.1 Scalar Conversion Tests

The scalar tests check that

```
mpir::scalar_cast<T_out>(x)
```

can be instantiated for representative input/output type pairs. For each pair, several sample values are tested:

```
0.0  
1.0  
-2.5  
1.0 / 3.0  
1.23456789012345
```

The result is checked at compile time with

```
static_assert(std::is_same_v<decltype(y), T_out>);
```

To verify that `scalar_cast<T_out>(x)` returns the requested output type `T_out`.

Both the input and the converted value are transformed to their approximate `double` representation with `approx_as_double` in order to compare the values with various `assert` and inspect the outputs.

3.3.2 Lossy Conversion Tests

It is verified that conversion from `double` to `CPFloat` indeed loses precision using a value such as 1.23456789012345.

This is only a useful sanity check that the conversion is really taking place.

3.3.3 Lossless Conversion Tests

It is also verified that conversion from a lower precision to a higher precision do not indeed lose information.

3.3.4 Vector Conversion Tests

Vector tests check the function

```
mpir::convert(hdnum::Vector<T_out>& out,  
              const hdnum::Vector<T_in>& in)
```

for a representative vector of length 5 and checks that each component of the output agrees with applying `scalar_cast<T_out>` to the corresponding input component.

The comparison is performed approximately by converting both values to `double` and using a small relative tolerance with the function

```
inline bool close(double a, double b, double atol = 0.0, double rtol = 1e-12)  
{  
    const double scale = std::max(std::abs(a), std::abs(b));  
    return std::abs(a - b) <= atol + rtol * scale;  
}
```

This function says whether the two double values are close enough to be considered equal. The tolerance is scaled by the max value of the inputs.

The test also includes the edge cases:

- vector size 0
- vector size 1

Additionally wrong-sized output vector is checked to verify whether an exception is thrown.

Finally the convenience function

```
mpir::convert_vector<T_out>(in)
```

is tested, by verifying the resulting type and converted values.

3.3.5 Matrix Conversion Tests

The matrix tests check the function

```
mpir::convert(hdnum::DenseMatrix<T_out>& out,  
              const hdnum::DenseMatrix<T_in>& in)
```

on a representative 5 x 5 matrix, by looping over the output values and comparing them against the result of applying `scalar_cast` to corresponding input entries.

The test includes the edge case of 1 x 1 matrix and a wrong-sized test.

The convenience function `mpir::convert_matrix<T_out>(in)` is also tested.

3.3.6 Representative Pair Types

1. Native $\leftrightarrow$ Native:
 - FP32 $\leftrightarrow$ FP64
2. Working precision $u \rightarrow$ low precision u_f :
 - FP64 $\rightarrow$ FP16 / bfloat16 / FP8
3. Low Precision $u_f \rightarrow$ working precision u :
 - FP16 / bfloat16 / FP8 $\rightarrow$ FP64
4. Working precision $u \rightarrow$ residual precision u_r
 - FP64 $\rightarrow$ FP128 / FP256
5. Residual precision $u_r \rightarrow$ working precision u
 - FP128 / FP256 $\rightarrow$ FP64
6. Low precision $u_f \leftrightarrow$ high residual/reference precision u_r
 - FP128 / FP256 $\leftrightarrow$ FP16 / bfloat16 / FP8
7. High precision $\leftrightarrow$ high precision
 - FP128 $\leftrightarrow$ FP256

Covering the main conversion direction required later by the algorithm:

- $u \rightarrow u_f$
- $u_f \rightarrow u$
- $u \rightarrow u_r$
- $u_r \rightarrow u$

The tests also include low to high-precision conversions not required directly by the algorithm but useful nevertheless.

4 Implementation of the Mixed IR Algorithm

4.1 Review of LU Decomposition

4.1.1 Partial Pivoting

In partial pivoting only rows are possibly permuted and the permutations are ‘stored’ in a matrix P . So we compute the decomposition:

$$LU = PA$$

This can be used to solve the matrix equation $Ax = b$ as follows:

$$\begin{aligned} Ax &= b \\ \Leftrightarrow (PA)x &= Pb \\ \Leftrightarrow L(Ux) &= Pb & (LU = PA, \text{assoc}) \\ \Rightarrow Ly &= Pb & (y := Ux) \\ \text{solve for } y &\text{ using forward subst} \\ \text{solve } Ux = y &\text{ for } x \text{ using backward subst} \end{aligned}$$

4.1.2 Full Pivoting

In full pivoting columns can be permuted in addition to the row permutation, improving backward stability of the algorithm, s.t. the following equation holds:

$$PAQ = LU$$

Here P , Q are the row and column permutation matrices. This decomposition can be used to solve the equation $Ax = b$ as follows:

$$\begin{aligned}
& Ax = b \\
& \Leftrightarrow PA(QQ^{-1})x = b \\
& \Leftrightarrow (PAQ)Q^{-1}x = b \\
& \Leftrightarrow L(UQ^{-1})x = b & (LU = PAQ) \\
& \Rightarrow Lz = b & (UQ^{-1} := z) \\
& \text{Solve for } z \text{ using forward subst.} \\
& \text{We have:} \\
& \Rightarrow Uy = z & (Q^{-1}x := y) \\
& \text{Solve for } y \text{ using backward subst.} \\
& \text{Compute } x = Qy
\end{aligned}$$

4.1.3 Implementation in HDNum

Solving $Ax = b$ using LU decomposition is implemented in HDnum as follows. Consider:

$$A = \begin{bmatrix} 2 & 1 & 7 \\ 8 & 8 & 33 \\ -4 & 10 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 15 \\ 73 \\ 12 \end{bmatrix}$$

We enter this as follows:

```

hdnum::Vector<hdnum::FP256> b(3);
b[0] = 15;
b[1] = 73;
b[2] = 12;

hdnum::DenseMatrix<hdnum::FP256> A(3, 3);
A[0][0] = 2;   A[0][1] = 1;   A[0][2] = 7;
A[1][0] = 8;   A[1][1] = 8;   A[1][2] = 33;
A[2][0] = -4;  A[2][1] = 10;  A[2][2] = 4;

```

We also need the vector x to hold the solution, and the vectors p and q to hold the row and column permutations:

```

hdnum::Vector<hdnum::FP256> x(3, 0.0); // holds the solution
hdnum::Vector<std::size_t> p(3);
hdnum::Vector<std::size_t> q(3);

```

Then we carry out the LU decomposition as follows:

```
hdnum::lr_fullpivot(A, p, q);
```

After this command A is modified and holds both L and U , while p and q hold the permutation information. We also have to permute b accordingly:

```
hdnum::permute_forward(p, b);
```

Then we solve the $Ly = b$ part of the equation (the forward substitution):

```
hdnum::solveL(A, b, b);
```

Note that we are overwriting b and storing the result directly in it to save space. Then, we finally complete the solution with:

```
hdnum::solveR(A, x, b);
```

Since we did full-pivoting, we should permute back the solution with:

```
hdnum::permute_backward(q, x);
```

Now x holds the solution.

4.2 The Mixed IR Algorithm

First we remember the mixed-ir algorithm:

1. Compute LU factorization $PAQ = LU$ in precision uf .
2. Solve $LU x_0 = P b$ in precision uf ;
Cast x_0 to precision u
3. While not converged:
 4. Compute $r_i = b - A x_i$ in precision ur (residual)
 5. cast r_i to precision u .
 6. Solve $LU d_i = P r_i$ in precision uf ;
Cast d_i to precision u
 7. update $x(i+1) = x_i + d_i$ in precision u

- Convergence condition $\frac{\|d_i\|}{\|x_i\|} < u$, where u is the unit roundoff of the **working precision**, or stop after a max number of iterations.

The c++ implementation of this algorithm is given as follows. First we define an options struct that holds the options that control max iterations and relative correction tolerance, i.e. the termination condition.

```
struct MixedIROptions {
    std::size_t max_iterations = 10;
    double rel_correction_tol = 0.0;
};
```

The default value `rel_correction_tol = 0.0` causes the algorithm to try to use the default unit roundoff of the data type used for the working precision u , as we'll see later in the algorithm.

Next we define a struct to hold the result, here the comments provide the explanations:

```
template<class T_work>
struct MixedIRResult {
    // Final approximate solution, stored in the working precision T_work.
    hdnum::Vector<T_work> x;

    // Number of refinement iterations actually performed.
    // This does not count the initial LU solve for x0.
    std::size_t iterations = 0;

    // True if the stopping criterion was reached before max_iterations.
    // False means the algorithm stopped because the iteration limit was reached.
    bool converged = false;

    // Last computed relative correction norm ||d_i|| / ||x_i||.
    // Useful for diagnostics, especially if converged == false.
    double final_rel_correction = 0.0;

    // History of relative correction norms, one entry per refinement iteration.
    // Useful for convergence diagnostics and later plotting.
    std::vector<double> rel_corrections;
};
```

We define a function that returns u of the templatized type T , if it's obtainable, or throws an error if not. (Mostly we will use `double` for u , for which it can be computed in compile time):

```

template<class T>
double default_unit_roundoff()
{
    static_assert(std::numeric_limits<T>::is_specialized,
        "default_unit_roundoff: no numeric_limits specialization for this type;"
        "please set_rel_correction_tol_explicitly");

    return 0.5 * static_cast<double>(std::numeric_limits<T>::epsilon());
}

```

Next a function that computes the 2-norm in double precision for a vector of a generic T type.

```

template<class T>
double norm2_as_double(const hdnum::Vector<T>& v)
{
    double sum = 0.0;

    for (std::size_t i = 0; i < v.size(); ++i) {
        const double vi = scalar_cast<double>(v[i]);
        sum += vi * vi;
    }

    return std::sqrt(sum);
}

```

Then we define a function that encapsulates the steps to solve an equation $Ax = b$ after LU decomposition is done. As input it expects the result of the LU decomposition, the permutation vectors p and q , and the vector b :

```

template<class T>
hdnum::Vector<T>
solve_with_lu_fullpivot(
    const hdnum::DenseMatrix<T>& LU,
    const hdnum::Vector<std::size_t>& p,
    const hdnum::Vector<std::size_t>& q,
    const hdnum::Vector<T>& b
)
{
    hdnum::Vector<T> y(b); // copy b
    hdnum::Vector<T> x(b.size()); // holds solution
}

```

```

    hdnum::permute_forward(p, y);
    hdnum::solveL(LU, y, y); // Solve L*y = P*b, in-place in y.
    hdnum::solveR(LU, x, y); // Solve U*x = y
    hdnum::permute_backward(q, x); // undo column permutation
    return x;
}

```

Then a function that performs the residual computation with the necessary type conversions. It expects a matrix A , vectors b and current solution x , all in working precision u , converts them to higher precisions, and returns the residual $r = Ax - b$. in higher residual precision u_r :

```

template<class T_residual, class T_work>
hdnum::Vector<T_residual> compute_residual(
    const hdnum::DenseMatrix<T_work>& A,
    const hdnum::Vector<T_work>& b,
    const hdnum::Vector<T_work>& x
)
{
    const std::size_t n = b.size();

    hdnum::DenseMatrix<T_residual> A_r(A.rowsize(), A.colsize());
    hdnum::Vector<T_residual> b_r(n);
    hdnum::Vector<T_residual> x_r(n);
    hdnum::Vector<T_residual> Ax_r(n);
    hdnum::Vector<T_residual> r_r(n);

    convert(A_r, A);
    convert(b_r, b);
    convert(x_r, x);

    A_r.mv(Ax_r, x_r);

    for (std::size_t i = 0; i < n; i++) {
        r_r[i] = b_r[i] - Ax_r[i];
    }

    return r_r;
}

```

Finally we provide the full algorithm, where the comments provide the necessary explanations:

```

template <class T_factor, class T_work, class T_residual>
MixedIRResult<T_work> mixed_ir(
    const hdnum::DenseMatrix<T_work>& A,
    const hdnum::Vector<T_work>& b,
    const MixedIROptions& options = {}
)
{
    if (A.rowsize() != A.colsize() || A.rowsize() == 0) {
        throw std::invalid_argument("mixed_ir: A must be square and non-empty");
    }

    if (A.rowsize() != b.size()) {
        throw std::invalid_argument("mixed_ir: A must be square and nonempty");
    }

    const std::size_t n = b.size();

    // if tol is default value 0.0, then use the unit roundoff of T_work
    // which is compute in compile time
    // otherwise use the value passed in the options.
    const double tol =
        options.rel_correction_tol > 0.0 ? options.rel_correction_tol
        : default_unit_roundoff<T_work>();

    // 1. convert A and b to factorization precision.
    hdnum::DenseMatrix<T_factor> A_f(n, n);
    hdnum::Vector<T_factor> b_f(n);

    convert(A_f, A);
    convert(b_f, b);

    // 2. Compute LU factorization in low precision
    //
    // lr_fullpivot overwrites A_f with the combined L / U factors
    hdnum::Vector<std::size_t> p(n);
    hdnum::Vector<std::size_t> q(n);

    hdnum::lr_fullpivot(A_f, p, q);

    // 3. Initial low-precision solve
    hdnum::Vector<T_factor> x0_f = solve_with_lu_fullpivot(A_f, p, q, b_f);

```

```

// 4. cast x0 to working precision
MixedIRResult<T_work> result;
result.x = hdnum::Vector<T_work>(n);
convert(result.x, x0_f);

// 5. refinement loop
for (std::size_t k = 0; k < options.max_iterations; k++) {
    // Compute residual  $r = b - A * x$  in residual precision.
    hdnum::Vector<T_residual> r_r = compute_residual<T_residual>(A, b, result.x);

    // cast residual to working precision
    hdnum::Vector<T_work> r_w(n);
    convert(r_w, r_r);

    // The triangular solve uses T_factor L and U factors,
    // so the right-hand side must also be in T_factor
    hdnum::Vector<T_factor> r_f(n);
    convert(r_f, r_w);

    // Solve correction equation  $A d = r$  using existing LU factors:
    hdnum::Vector<T_factor> d_f = solve_with_lu_fullpivot(A_f, p, q, r_f);

    // Cast correction back to working precision
    hdnum::Vector<T_work> d_w(n);
    convert(d_w, d_f);

    const double norm_d = norm2_as_double(d_w);
    const double norm_x = norm2_as_double(result.x);
    const double denominator = (norm_x > 0.0) ? norm_x : 1.0;

    const double rel_corr = norm_d / denominator;

    result.rel_corrections.push_back(rel_corr);
    result.final_rel_correction = rel_corr;

    // update in working precision
    for (std::size_t i = 0; i < n; i++) {
        result.x[i] += d_w[i];
    }

    result.iterations = k + 1;
}

```

```

        if (norm_d <= tol * norm_x) {
            result.converged = true;
            break;
        }

    }

    return result;
}

```

4.3 Unit Roundoff u

For a floating point format with t significand bits the unit round off is defined as

$$\begin{aligned}
 u &= \frac{1}{2}\beta^{1-t} \\
 &= 2^{-t} \quad (\beta = 2)
 \end{aligned}$$

hdnum's cfloat wrapper's are:

```

using FP16 = hdnum::CPFloat<11,5>;
using bfloat16 = hdnum::CPFloat<8,8>;
using FP8 = hdnum::CPFloat<4,4>;

```

Where the first template parameters are the significand (or the mantisse) bits.

and the FP wrappers:

```

using FP128 = FP<64>;
using FP256 = FP<192>;
using FP512 = FP<448>;
using FP1024 = FP<960>;

```

where the template parameters are the mantisse bits.

From here we can construct the following table summarizing the unit roundoff u for the various precisions:

HDNUM type	Actual HDNUM alias	Significand precision (t)	Unit roundoff $u = 2^{-t}$
FP8	CPFloat<4,4>	4	$2^{-4} \approx 6.25 \cdot 10^{-2}$
bfloat16	CPFloat<8,8>	8	$2^{-8} \approx 3.91 \cdot 10^{-3}$
FP16	CPFloat<11,5>	11	$2^{-11} \approx 4.88 \cdot 10^{-4}$
FP32	float	24	$2^{-24} \approx 5.96 \cdot 10^{-8}$
FP64	double	53	$2^{-53} \approx 1.11 \cdot 10^{-16}$
FP128	FP<64>	64	$2^{-64} \approx 5.42 \cdot 10^{-20}$
FP256	FP<192>	192	$2^{-192} \approx 1.59 \cdot 10^{-58}$
FP512	FP<448>	448	$2^{-448} \approx 1.38 \cdot 10^{-135}$
FP1024	FP<960>	960	$2^{-960} \approx 1.03 \cdot 10^{-289}$

4.3.1 unit roundoff traits

Originally we had the function

```
template<class T>
double default_unit_roundoff()
{
    static_assert(std::numeric_limits<T>::is_specialized,
        "default_unit_roundoff: no numeric_limits specialization for this type;"
        "please set_rel_correction_tol_explicitly");

    return 0.5 * static_cast<double>(std::numeric_limits<T>::epsilon());
}
```

This works only for data types `float` and `double`. We improve this using type traits, s.t. unit roundoffs are returned for other floating point formats at compile time:

```
#pragma once

#include <limits>
#include <type_traits>

#include "hdnum.hh"

namespace mpir {

constexpr double pow2_neg(int p)
{
    double x = 1.0;
```

```

    for (int i = 0; i < p; ++i) {
        x *= 0.5;
    }
    return x;
}

namespace detail {

template<class>
struct unit_roundoff_always_false : std::false_type {};

} // namespace detail

template<class T>
struct unit_roundoff_traits {
    static_assert(detail::unit_roundoff_always_false<T>::value,
        "unit_roundoff_traits: unsupported scalar type");
};

template<>
struct unit_roundoff_traits<hdnum::FP32> {
    static constexpr double value =
        0.5 * static_cast<double>(std::numeric_limits<hdnum::FP32>::epsilon());
};

template<>
struct unit_roundoff_traits<hdnum::FP64> {
    static constexpr double value =
        0.5 * static_cast<double>(std::numeric_limits<hdnum::FP64>::epsilon());
};

#ifdef HDNUM_HAS_CPFLOAT
template<int m, int e>
struct unit_roundoff_traits<hdnum::CPFfloat<m,e>> {
    static constexpr double value = pow2_neg(m);
};
#endif

#ifdef HDNUM_HAS_GMP
template<int m>
struct unit_roundoff_traits<hdnum::FP<m>> {
    static constexpr double value = pow2_neg(m);
};

```



```
};  
#endif  
  
template<class T>  
constexpr double default_unit_roundoff()  
{  
    return unit_roundoff_traits<T>::value;  
}  
  
} // namespace mpir
```

Now mixed ir algorithm will work for all fp formats without having to specify tol manually in the options struct.

5 Implementing the Experiments

We want to plot the **forward error** and the **backward error** as function of the refinement step i .

forward error:

$$\text{forward error}_i = \frac{\|x_i - x_{\text{true}}\|}{\|x_{\text{true}}\|}$$

Backward error:

$$\eta_i = \frac{\|b - Ax_i\|}{\|A\| \|x_i\| + \|b\|}$$

in `mixed_ir.hpp` the result struct currently only stores `rel_correction` vector, that records

$$\frac{\|d_i\|}{\|x_i\|}$$

We have to modify `mixed_ir` to store the iterates x_i , so that we can compute forward and backward errors. We add the following member variable to the `MixedIRResult` struct:

```
std::vector<hdnum::Vector<T_work>> iterates;
```

We append the solutions to the iterates vector before the loop and after each loop iteration with the line:

```
if (options.store_iterates) {  
    result.iterates.push_back(result.x);  
}
```

5.1 Temporary Matrix Generator

We start with a simple controlled family, a rotated SPD matrix:

$$A = Q\Lambda Q^T$$

where Λ has eigenvalues $1/\sqrt{\kappa} \dots \sqrt{\kappa}$. Then

$$\kappa_2(A) = \kappa$$

This family of matrices give a known true (exact) solution:

$$x_{\text{true}} = (1, \dots, 1)^T$$

So it can be used temporarily to test the pipeline.

The implementation is in

`code/include/test_matrices.hpp`

5.2 Matrix Generator Supervisor Version

The temporary generator constructs

$$A = Q_{\text{block}}\Lambda Q_{\text{block}}^T$$

where Q_{block} consists only of independent 2×2 rotations. Consequently A is block diagonal:

$$A = \begin{pmatrix} A_1 & & & 0 \\ & A_2 & & \\ & & \ddots & \\ 0 & & & A_m \end{pmatrix}, \quad A_i \in \mathbb{R}^{2 \times 2}.$$

Even though the entire block-diagonal matrix had condition number κ , the individual 2×2 blocks is only approximately:

$$\kappa^{1/(n-1)}$$

Supervisors `randspd` constructs

$$A = Q\Lambda Q^T$$

where Q is a dense random orthogonal matrix. Consequently, A is generally dense, and every variable is coupled to every other variable, producing a genuinely ill-conditioned system.

5.2.1 Nonsymmetric Generator

Iterative refinement with LU is not restricted to SPD matrices, so `randsvd` is also important. The wrapper

```
template<class T>
LinearSystem<T>
make_random_svd_problem(std::size_t n,
                        double kappa,
                        unsigned int seed_u = 42,
                        unsigned int seed_v = 137)
{
    hdnum::DenseMatrix<T> A(n, n);
    hdnum::Vector<T> x_true(n);
    hdnum::Vector<T> b(n);

    hdnum::randsvd(
        A,
        scalar_cast<T>(kappa),
        seed_u,
        seed_v
    );

    for (std::size_t i = 0; i < n; ++i) {
        x_true[i] = T(1);
    }

    A.mv(b, x_true);

    return LinearSystem<T>{
        A,
        b,
        x_true,
        kappa
    };
}
```

produces

$$A = U\Sigma V^T$$

with singular values $1, \dots, 1/\kappa$. It is nonsymmetric and a more general test of the iterative LU algorithm.

5.3 Configurable test-problem Construction

The test-matrix infrastructure contained in ‘test_matrices.hpp’ supports three different ways of constructing the right-hand side b and solution of

$$Ax = b$$

The matrix generator and the right-hand-side strategy are separated:

1. A matrix generator constructs A
2. A shared `complete_problem` function constructs b .
3. The reference solution is computed in a separate, higher precision.

5.3.1 Three Right-hand-side Modes

There are three available strategies represented by:

```
enum class RightHandSideMode {  
    ones_solution,  
    random_sign_solution,  
    random_normal_rhs  
};
```

- `ones_solution`:

$$x = (1, \dots, 1)^T$$
$$b = \text{fl}(Ax)$$

- `random_sign_solution`:

Construct a reproducible random vector with

$$x_i \in \{-1, 1\}$$

and form

$$b = \text{fl}(Ax)$$

- `random_normal_rhs`:

Here we generate the right-hand-side **directly** (randomly):

$$b_i \sim \mathcal{N}(0, 1)$$

In all these cases the reference (true) solution is obtained by solving

$$Ax_{\text{ref}} = b$$

in a high precision (called measure or reference precision).

5.3.2 Test-problem Options

The choices and random seeds are collected in:

```
struct TestProblemOptions {
    RightHandSideMode rhs_mode =
        RightHandSideMode::ones_solution;

    unsigned int matrix_seed_u = 42;
    unsigned int matrix_seed_v = 137;
    unsigned int vector_seed = 2718;

    double rotation_theta = 0.3;
};
```

- `rhs_mode`: selects how b is constructed
- `matrix_seed_u`: seed for the first random orthogonal matrix (used by `randspd`)
- `matrix_seed_v`: seed for the second random orthogonal matrix (used by `randsvd`)
- `vector_seed`: seed for the random-sign solution or random-normal right-hand side.
- `rotation_theta`: rotation angle used by the structured rotated-SPD generator.

The default mode is `ones_solution`.

5.3.3 Linear-system Representation

The system stores the problem and reference solution (potentially in different precisions):

```
template<class T_data, class T_reference = T_data>
struct LinearSystem {
    hdnum::DenseMatrix<T_data> A;
    hdnum::Vector<T_data> b;
    hdnum::Vector<T_reference> x_true;
    double kappa;
};
```

- `T_data`: the precision in which A and b are stored, normally `T_work`.
- `T_reference`: precision used for the reference solution, normally `T_measure`.

5.3.4 Shared `complete_problem` function

We have three matrix generators,

- `make_rotated_spd_problem`
- `make_random_spd_problem`
- `make_random_svd_problem`

each created only A and delegates the remaining work to `create_problem`, the signature:

```
template<class T_data, class T_reference>
LinearSystem<T_data, T_reference>
complete_problem(
    hdnum::DenseMatrix<T_data> A,
    double kappa,
    const TestProblemOptions& options);
```

The logical structure is:

```
if (options.rhs_mode ==
    RightHandSideMode::ones_solution) {

    // Construct  $x = (1, \dots, 1)$ .
    // Compute  $b = A*x$  in T_data.
}
else if (options.rhs_mode ==
    RightHandSideMode::random_sign_solution) {
```

```

    // Construct reproducible x_i in {-1,1}.
    // Compute b = A*x in T_data.
}
else {
    // Construct reproducible b_i ~ N(0,1)
    // directly in T_data.
}

```

After b is constructed the reference solution is computed:

```

auto x_reference =
    high_precision_solve<T_reference>(A, b);

```

i The reference solution is recomputed

Even in the first two modes, even if we explicitly construct x , we recompute

$$x_{\text{true}} = \text{solve}_{T_{\text{ref}}}(A, b)$$

because the matrix-vector product

$$b = \text{fl}(Ax_{\text{constructed}})$$

is founded. Therefore, the exact solution of the stored pair (A, b) can differ slightly from $x_{\text{constructed}}$.

5.3.5 High-precision reference solve

The helper

```

template<class T_reference, class T_data>
hdnum::Vector<T_reference>
high_precision_solve(
    const hdnum::DenseMatrix<T_data>& A,
    const hdnum::Vector<T_data>& b);

```

contained in `reference_solve.hpp` computes a reference solution using `hdnums` LU-solver functions in `T_reference` precision.

For example, `T_reference = hdnum::FP256` provides a sufficiently accurate reference solution.

5.3.6 Matrix-generator interfaces

As mentioned there are three generators, all with compatible interfaces:

```
template<class T_data, class T_reference = T_data>
LinearSystem<T_data, T_reference>
make_rotated_spd_problem(
    std::size_t n,
    double kappa,
    const TestProblemOptions& options = {});
```

```
template<class T_data, class T_reference = T_data>
LinearSystem<T_data, T_reference>
make_random_spd_problem(
    std::size_t n,
    double kappa,
    const TestProblemOptions& options = {});
```

```
template<class T_data, class T_reference = T_data>
LinearSystem<T_data, T_reference>
make_random_svd_problem(
    std::size_t n,
    double kappa,
    const TestProblemOptions& options = {});
```

Each generator

1. constructs its particular matrix A
2. calls `complete_problem`
3. returns `LinearSystem<T_data, T_reference>` struct.

Which enables plugging in different matrix families in an experiment.

5.3.7 Usage Examples

Define the precisions:

```
using T_work = hdnum::FP64;
using T_measure = hdnum::FP256;
```

- all-ones x :

```

mpir::TestProblemOptions problem_options;

problem_options.rhs_mode =
    mpir::RightHandSideMode::ones_solution;

auto problem =
    mpir::make_random_spd_problem<T_work, T_measure>(
        n,
        kappa,
        problem_options
    );

```

- random-sign ones x :

```

mpir::TestProblemOptions problem_options;

problem_options.rhs_mode =
    mpir::RightHandSideMode::random_sign_solution;

problem_options.vector_seed = 2718;

auto problem =
    mpir::make_random_spd_problem<T_work, T_measure>(
        n,
        kappa,
        problem_options
    );

```

- random-normal b :

```

mpir::TestProblemOptions problem_options;

problem_options.rhs_mode =
    mpir::RightHandSideMode::random_normal_rhs;

problem_options.vector_seed = 1;

auto problem =
    mpir::make_random_svd_problem<T_work, T_measure>(
        n,
        kappa,
        problem_options
    );

```

only the generator call needs to change, the resulting problems are used identically:

```
auto result =
    mpir::mixed_ir<T_factor, T_work, T_residual>(
        problem.A,
        problem.b,
        ir_options
    );
```

5.4 Error Metrics Helper

We implement functions to compute

- $\|\cdot\|_\infty$ for matrices and vectors,
- relative forward error with the infinity norm
- relative backward error with the infinity norm

contained in the file

`code/include/error_metrics.hpp`

5.5 First Convergence Test

We test our pipeline with a first test that uses the precision tripple:

- u_f : FP16
- u : FP64
- u_r : FP128

For matrices of size 100×100 with κ values: 1, 10, 100, 1000, 10000.

The test writes for each matrix the convergence history to csv file. Specifically it writes the values:

- kappa,
- iteration,
- relative forward error
- relative backward error
- relative correction

in a single lines for each iteration step, for each matrix. This gives us the convergence histories.

It is implemented in the file

`code/experiments/exp_convergence.cc`

Following was the result of the experiment:

```
kappa,iteration,forward_error_inf,backward_error_inf,rel_correction
1,0,0,0,0
1,1,0,0,0
10,0,0.000976562,0.000441793,0
10,1,9.53674e-07,4.31651e-07,0.00047591
10,2,5.96046e-08,5.1214e-09,3.1641e-07
10,3,5.96046e-08,5.1214e-09,1.68587e-08
10,4,5.96046e-08,5.1214e-09,1.68587e-08
10,5,5.96046e-08,5.1214e-09,1.68587e-08
10,6,5.96046e-08,5.1214e-09,1.68587e-08
10,7,5.96046e-08,5.1214e-09,1.68587e-08
10,8,5.96046e-08,5.1214e-09,1.68587e-08
10,9,5.96046e-08,5.1214e-09,1.68587e-08
10,10,5.96046e-08,5.1214e-09,1.68587e-08
10,11,5.96046e-08,5.1214e-09,1.68587e-08
10,12,5.96046e-08,5.1214e-09,1.68587e-08
10,13,5.96046e-08,5.1214e-09,1.68587e-08
10,14,5.96046e-08,5.1214e-09,1.68587e-08
10,15,5.96046e-08,5.1214e-09,1.68587e-08
10,16,5.96046e-08,5.1214e-09,1.68587e-08
10,17,5.96046e-08,5.1214e-09,1.68587e-08
10,18,5.96046e-08,5.1214e-09,1.68587e-08
10,19,5.96046e-08,5.1214e-09,1.68587e-08
10,20,5.96046e-08,5.1214e-09,1.68587e-08
100,0,0.000976562,0.000481752,0
100,1,9.53674e-07,4.70694e-07,0.000444832
100,2,1.78814e-07,1.42242e-09,2.52178e-07
100,3,1.78814e-07,1.42242e-09,0
1000,0,0.000976562,0.000460731,0
1000,1,9.53674e-07,3.07722e-07,0.000591975
1000,2,8.9407e-07,6.88472e-10,4.02364e-07
1000,3,8.9407e-07,5.65293e-10,1.3328e-08
1000,4,8.9407e-07,5.65293e-10,1.19209e-08
```

```

1000,5,8.9407e-07,5.65293e-10,1.19209e-08
1000,6,8.9407e-07,5.65293e-10,1.19209e-08
1000,7,8.9407e-07,5.65293e-10,1.19209e-08
1000,8,8.9407e-07,5.65293e-10,1.19209e-08
1000,9,8.9407e-07,5.65293e-10,1.19209e-08
1000,10,8.9407e-07,5.65293e-10,1.19209e-08
1000,11,8.9407e-07,5.65293e-10,1.19209e-08
1000,12,8.9407e-07,5.65293e-10,1.19209e-08
1000,13,8.9407e-07,5.65293e-10,1.19209e-08
1000,14,8.9407e-07,5.65293e-10,1.19209e-08
1000,15,8.9407e-07,5.65293e-10,1.19209e-08
1000,16,8.9407e-07,5.65293e-10,1.19209e-08
1000,17,8.9407e-07,5.65293e-10,1.19209e-08
1000,18,8.9407e-07,5.65293e-10,1.19209e-08
1000,19,8.9407e-07,5.65293e-10,1.19209e-08
1000,20,8.9407e-07,5.65293e-10,1.19209e-08
10000,0,0.000976562,0.000375237,0
10000,1,3.33786e-06,4.73515e-08,0.000488379
10000,2,9.53674e-07,3.8223e-10,5.28939e-07
10000,3,9.53674e-07,1.29324e-10,1.03238e-08
10000,4,9.53674e-07,1.29324e-10,0

```

5.6 Plotting Convergence Experiments

The script

```
code/scripts/plot_convergence.py
```

generates convergence-history and summary plots from one mixed-precision iterative-refinement CSV file.

It is designed to be reusable for different:

- matrix generators;
- right-hand-side or solution-generation modes;
- precision combinations.

The input CSV is selected using a command-line argument, so a separate plotting script is not needed for each experiment.

5.6.1 Expected CSV format

The CSV file must contain the columns:

```
kappa
iteration
forward_error_inf
backward_error_inf
rel_correction
converged
total_iterations
final_rel_correction
```

Each row describes one iterate for one condition number.

The usual relationship is:

```
iteration 0 → initial solution x0
iteration 1 → solution x1 after the first correction
iteration 2 → solution x2 after the second correction
...
```

5.6.2 Generated plots

For every input CSV, the script generates six plots.

Convergence histories

1. Relative forward error against refinement iteration.
2. Normwise backward error against refinement iteration.
3. Relative correction norm against refinement iteration.

Each condition number is displayed as a separate curve.

Summary plots

4. Final forward and backward errors against condition number.
5. Best forward and backward errors attained at any iteration.
6. Total refinement iterations against condition number.

The best-attained-error plot is useful when an iteration initially improves and later deteriorates. For example, a difficult ($=10000$) problem may attain its best forward error around iteration 7 but have a worse result at iteration 20.

The script also prints a numerical summary containing:

- condition number;
- convergence flag;
- total iterations;
- final forward error;
- final backward error;
- final relative correction;
- best forward and backward errors;
- iterations at which the best errors occurred.

5.6.3 Logarithmic plots and zero values

The error histories use logarithmic vertical axes. Since zero cannot be displayed on a logarithmic scale, the script replaces nonpositive plotting values with NaN.

This affects only plotting. The original CSV data is not modified.

Consequently, a curve or point with an exactly zero error may disappear from a logarithmic plot.

5.6.4 Command-line interface

The general invocation is:

```
python code/scripts/plot_convergence.py CSV_FILE \  
    --label "Description of experiment"
```

If the terminal is already inside `code/`, use:

```
python scripts/plot_convergence.py CSV_FILE \  
    --label "Description of experiment"
```

The script first checks whether `CSV_FILE` is a direct path. If it is not found there, it looks under:

```
code/results/raw/
```

The plots are written to:

```
code/results/plots/
```

5.6.5 Default behavior

The CSV argument is optional. If no filename is provided, the script uses:

```
convergence_fp16_fp64_fp128.csv
```

Therefore:

```
python code/scripts/plot_convergence.py
```

is equivalent to:

```
python code/scripts/plot_convergence.py \  
    convergence_fp16_fp64_fp128.csv
```

5.6.6 Output tags

The output tag is normally derived automatically from the CSV filename.

For example, from:

```
convergence_random_spd_fp16_fp64_fp128.csv
```

the script removes the initial `convergence_` and obtains:

```
random_spd_fp16_fp64_fp128
```

It then generates filenames such as:

```
convergence_forward_error_random_spd_fp16_fp64_fp128.png  
convergence_backward_error_random_spd_fp16_fp64_fp128.png  
convergence_rel_correction_random_spd_fp16_fp64_fp128.png  
summary_final_errors_random_spd_fp16_fp64_fp128.png  
summary_best_errors_random_spd_fp16_fp64_fp128.png  
summary_iterations_random_spd_fp16_fp64_fp128.png
```


A custom output tag can be supplied with:

```
--tag custom_name
```

For example:

```
python code/scripts/plot_convergence.py \  
    convergence_random_spd_fp16_fp64_fp128.csv \  
    --tag random_spd_test
```

5.6.7 Descriptive plot labels

The optional `--label` argument is added to plot titles:

```
--label "Random dense SPD"
```

This produces titles such as:

```
Random dense SPD: relative forward error, infinity norm vs. refinement iteration
```

The label does not affect the output filenames.

5.6.8 Matrix-family examples

Block-rotated SPD

```
python code/scripts/plot_convergence.py \  
    convergence_fp16_fp64_fp128.csv \  
    --label "Block-rotated SPD"
```

Random dense SPD

```
python code/scripts/plot_convergence.py \  
    convergence_random_spd_fp16_fp64_fp128.csv \  
    --label "Random dense SPD"
```

Random SVD

```
python code/scripts/plot_convergence.py \  
    convergence_random_svd_fp16_fp64_fp128.csv \  
    --label "Random SVD"
```

5.6.9 Encoding the solution or RHS mode

The test-problem infrastructure supports three modes:

```
RightHandSideMode::ones_solution  
RightHandSideMode::random_sign_solution  
RightHandSideMode::random_normal_rhs
```

These correspond to:

- construct $(x=(1,\dots,1)^T)$ and form $(b=Ax)$;
- construct random $(x_i \in [-1,1])$ and form $(b=Ax)$;
- generate $(b_i \sim N(0,1))$ and compute a high-precision reference solution.

The third mode uses a normal distribution, not a uniform distribution.

It is sensible to encode this choice in the CSV filename. A consistent naming scheme is:

```
convergence_<matrix-family>_<rhs-mode>_<factor>_<work>_<residual>.csv
```

For example:

```
convergence_random_spd_x_ones_fp16_fp64_fp128.csv  
convergence_random_spd_x_random_signs_fp16_fp64_fp128.csv  
convergence_random_spd_b_random_normal_fp16_fp64_fp128.csv
```

The plotting script automatically retains the RHS mode as part of the output tag. No code change is needed.

5.6.10 Examples for the three modes

All-ones solution

```
python code/scripts/plot_convergence.py \  
    convergence_random_spd_x_ones_fp16_fp64_fp128.csv \  
    --label "Random SPD, ones solution"
```

This produces files such as:

```
convergence_forward_error_random_spd_x_ones_fp16_fp64_fp128.png
```

Random-sign solution

```
python code/scripts/plot_convergence.py \  
    convergence_random_spd_x_random_signs_fp16_fp64_fp128.csv \  
    --label "Random SPD, random-sign solution"
```

This produces files such as:

```
convergence_forward_error_random_spd_x_random_signs_fp16_fp64_fp128.png
```

Random-normal right-hand side

```
python code/scripts/plot_convergence.py \  
    convergence_random_spd_b_random_normal_fp16_fp64_fp128.csv \  
    --label "Random SPD, normally distributed right-hand side"
```

This produces files such as:

```
convergence_forward_error_random_spd_b_random_normal_fp16_fp64_fp128.png
```

5.6.11 What the script detects automatically

The script automatically derives an output tag from the filename. It does not semantically interpret components such as:

```
random_spd
x_ones
x_random_signs
b_random_normal
fp16_fp64_fp128
```

Therefore:

- output filenames are generated correctly automatically;
- the same plots are generated for every mode;
- a human-readable title should currently be supplied through `--label`.

Automatic semantic interpretation could be added later, but it is not necessary for the current workflow.

5.6.12 Recording metadata inside the CSV

Encoding the experiment configuration in the filename is useful, but the CSV should eventually also record metadata such as:

```
matrix_family
rhs_mode
matrix_seed_u
matrix_seed_v
vector_seed
```

For example:

```
matrix_family,rhs_mode,matrix_seed_u,matrix_seed_v,vector_seed,kappa,...
random_spd,x_random_signs,42,137,2718,10,...
```

This makes the experiment self-describing even if the file is renamed.

5.6.13 Later comparison plots

`plot_convergence.py` plots one CSV dataset at a time. It should remain focused on that task.

A later script, for example

```
code/scripts/plot_compare_generators.py
```

or

```
code/scripts/plot_compare_rhs_modes.py
```

can place multiple experiments on the same figures.

Possible comparisons include:

- block-rotated SPD versus random SPD versus random SVD;
- all-ones versus random-sign solutions;
- prescribed (x) versus random-normal (b);
- different precision triples;
- different random seeds.

Thus, the intended organization is:

```
plot_convergence.py
    → detailed plots for one experiment

plot_compare_generators.py
    → compare matrix families

plot_compare_rhs_modes.py
    → compare solution/RHS-generation strategies
```

6 Report

6.1 This is some title